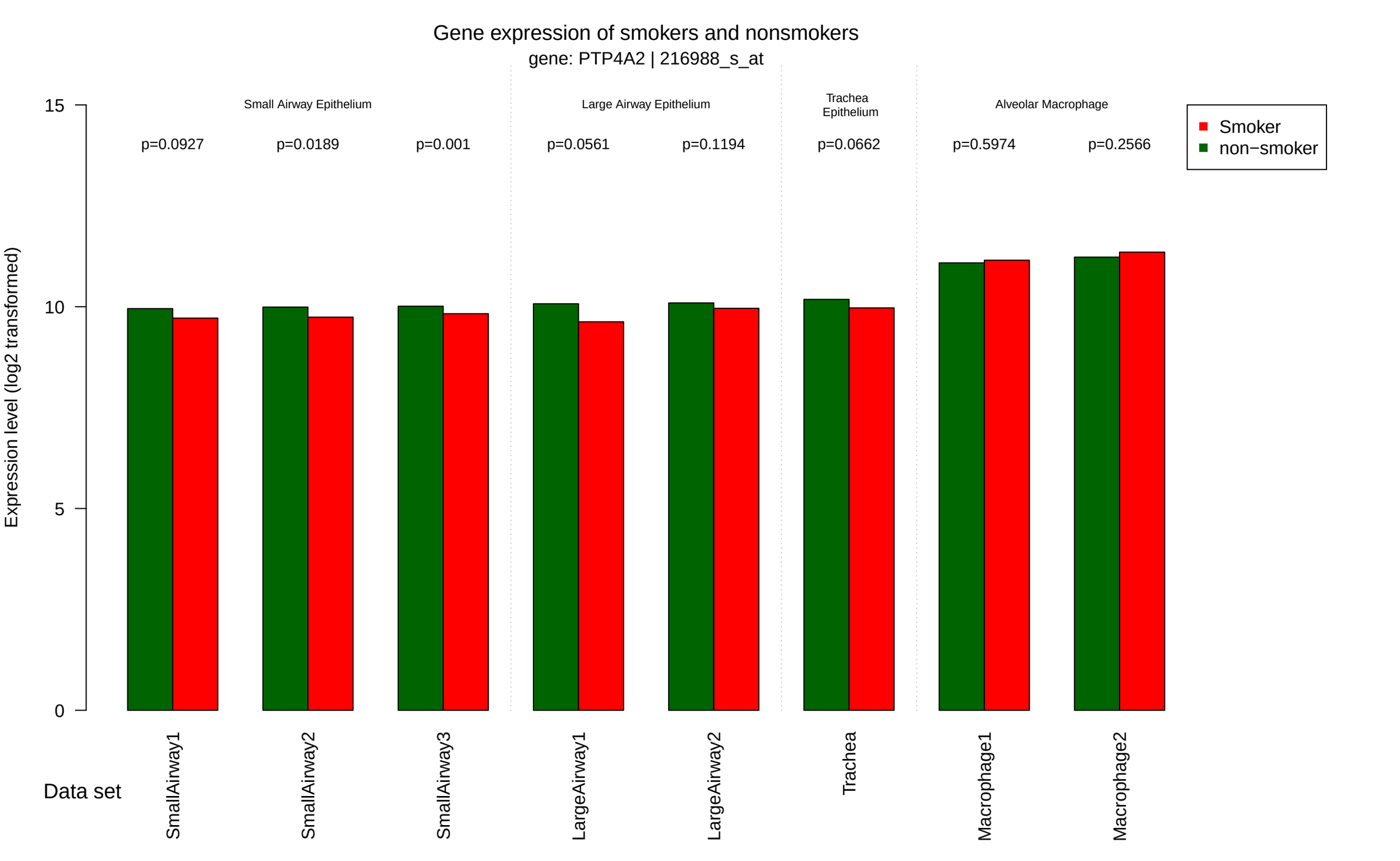
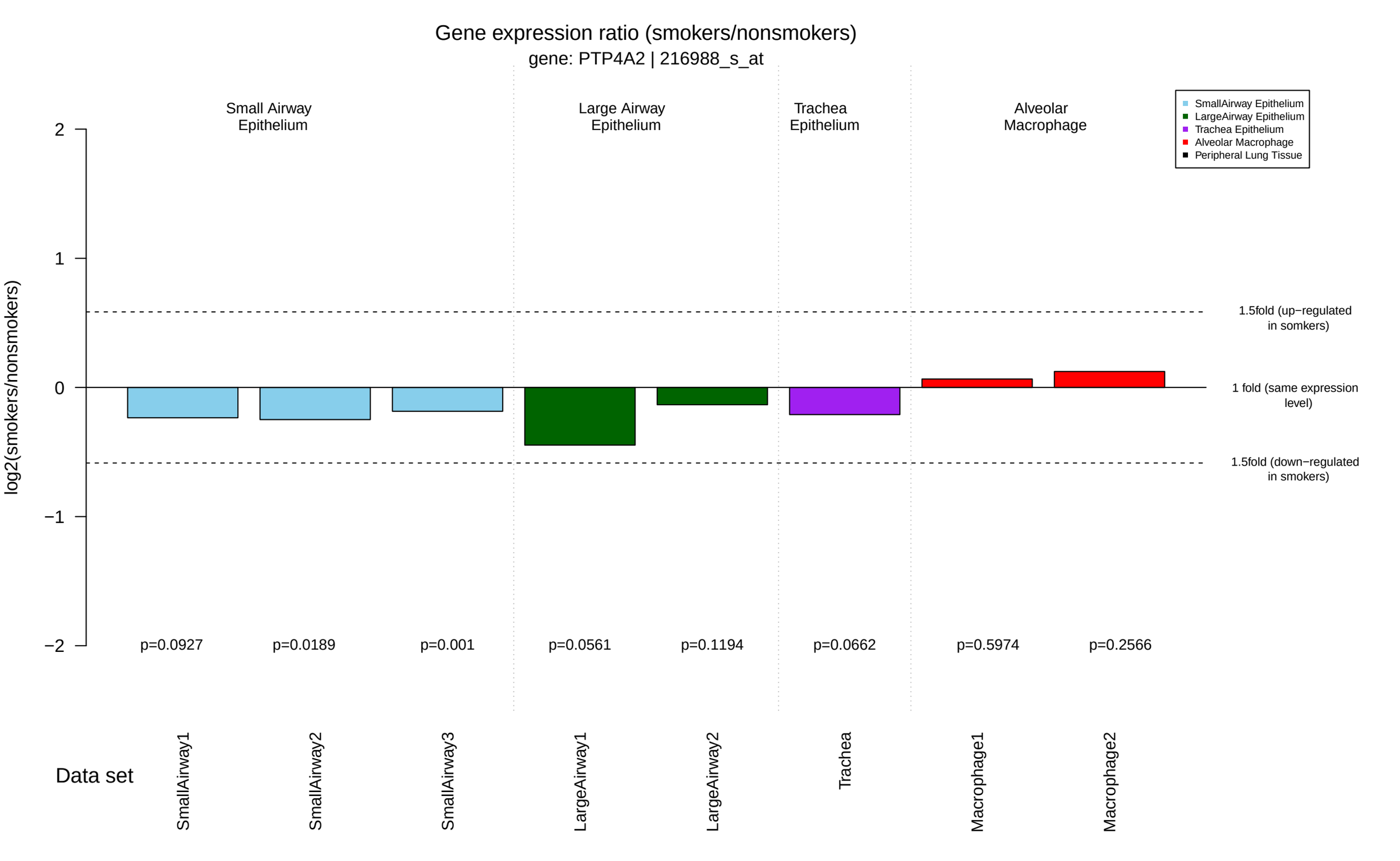
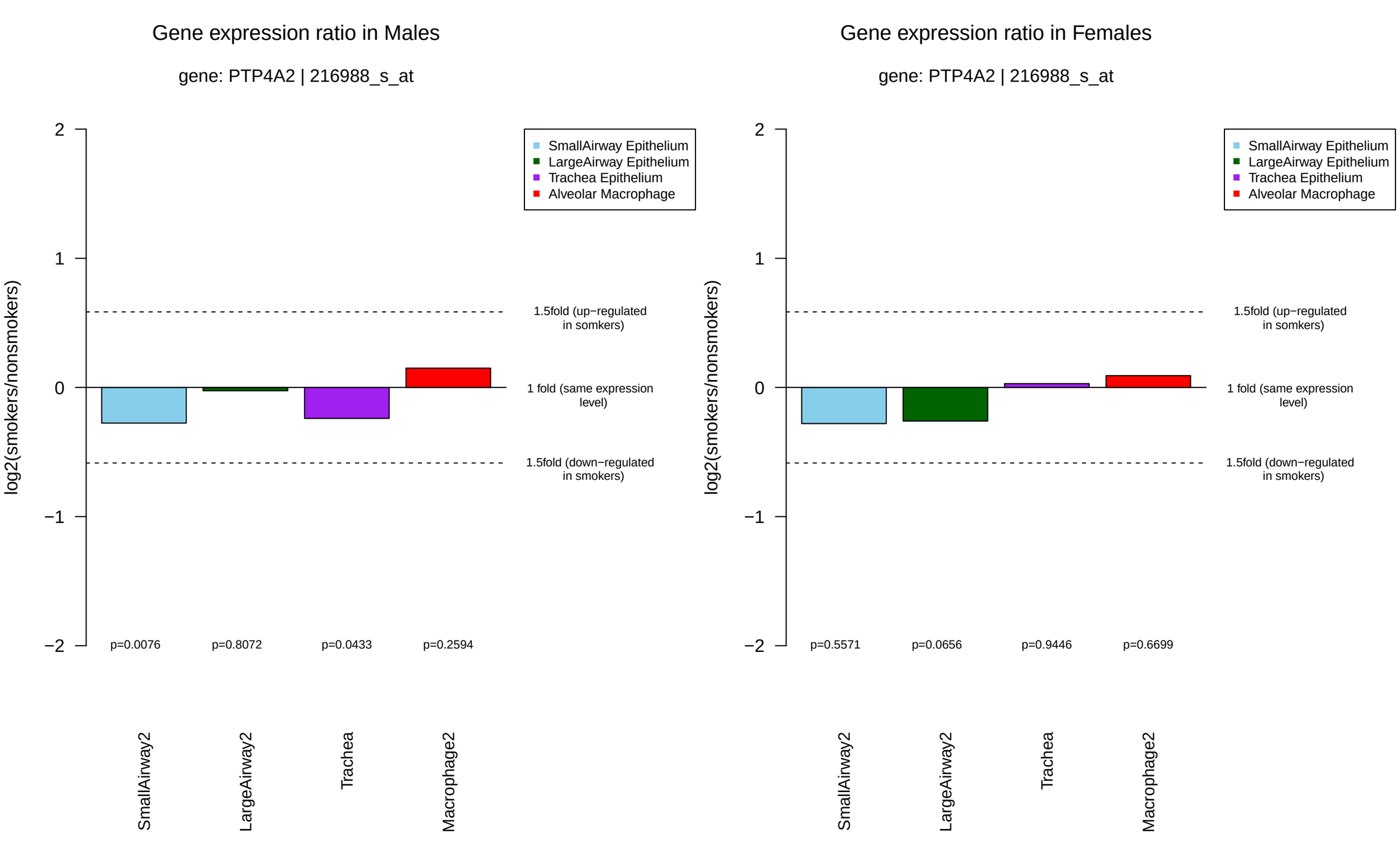
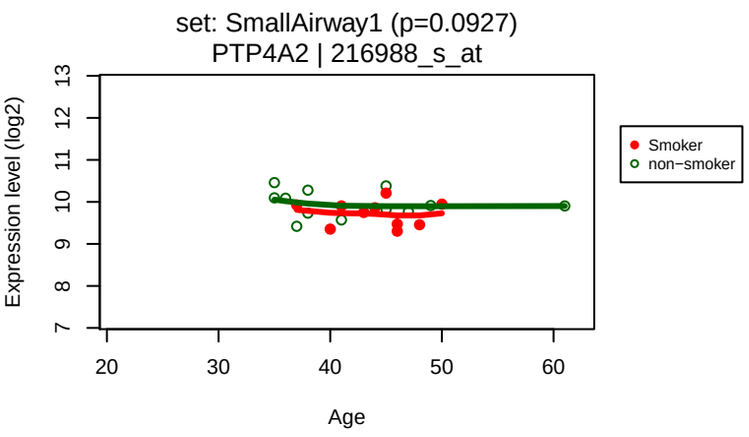
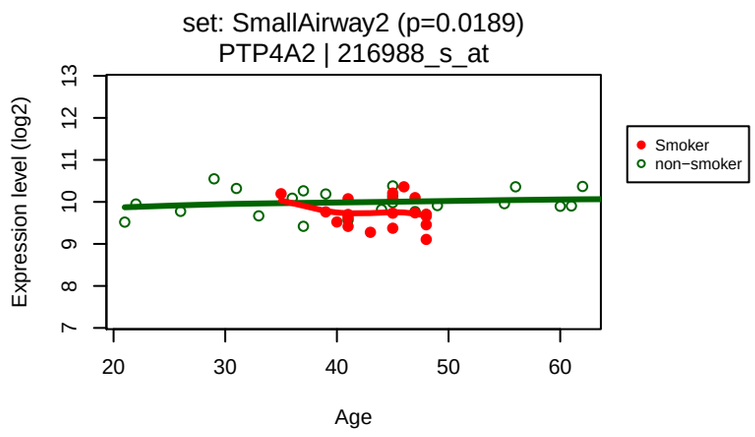
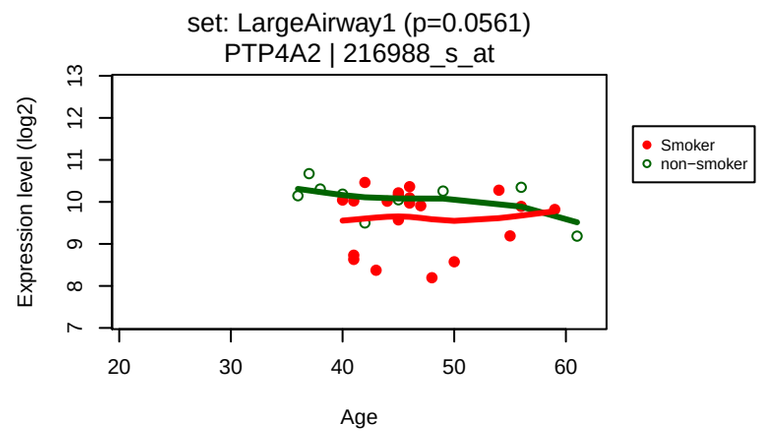

Gene expression of smokers and nonsmokers
gene: PTP4A2 | 216988_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0927
p=0.0189
p=0.001
p=0.0561
p=0.1194
p=0.0662
p=0.5974
p=0.2566
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

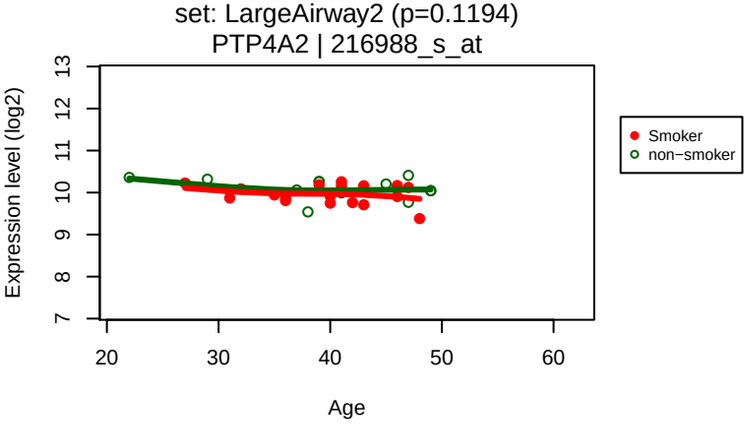

Gene expression ratio (smokers/nonsmokers)
gene: PTP4A2 | 216988_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0927
p=0.0189
p=0.001
p=0.0561
p=0.1194
p=0.0662
p=0.5974
p=0.2566
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

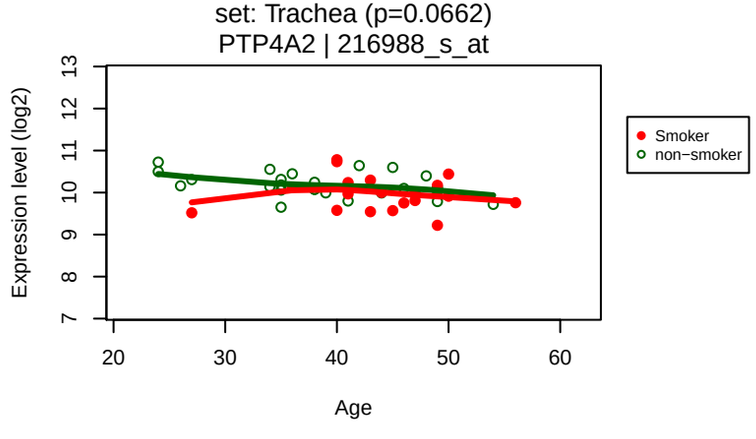

Gene expression ratio in Males
gene: PTP4A2 | 216988_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0076
p=0.8072
p=0.0433
p=0.2594
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTP4A2 | 216988_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5571
p=0.0656
p=0.9446
p=0.6699
SmallAirway2
LargeAirway2
Trachea
Macrophage2

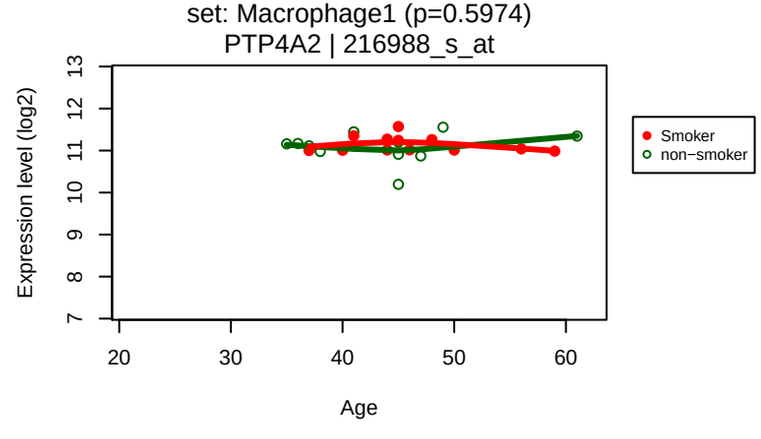

set: SmallAirway1 (p=0.0927)
PTP4A2 | 216988_s_at
Expression level (log2)
Age
Smoker
non−smoker

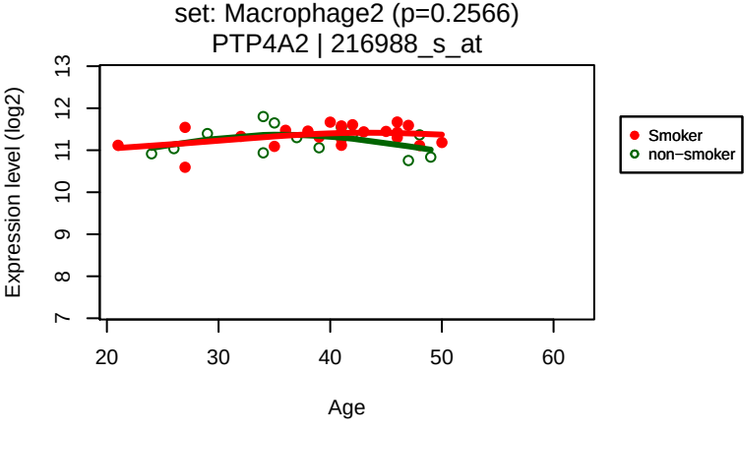

set: SmallAirway2 (p=0.0189)
PTP4A2 | 216988_s_at
Expression level (log2)
Age
Smoker
non−smoker

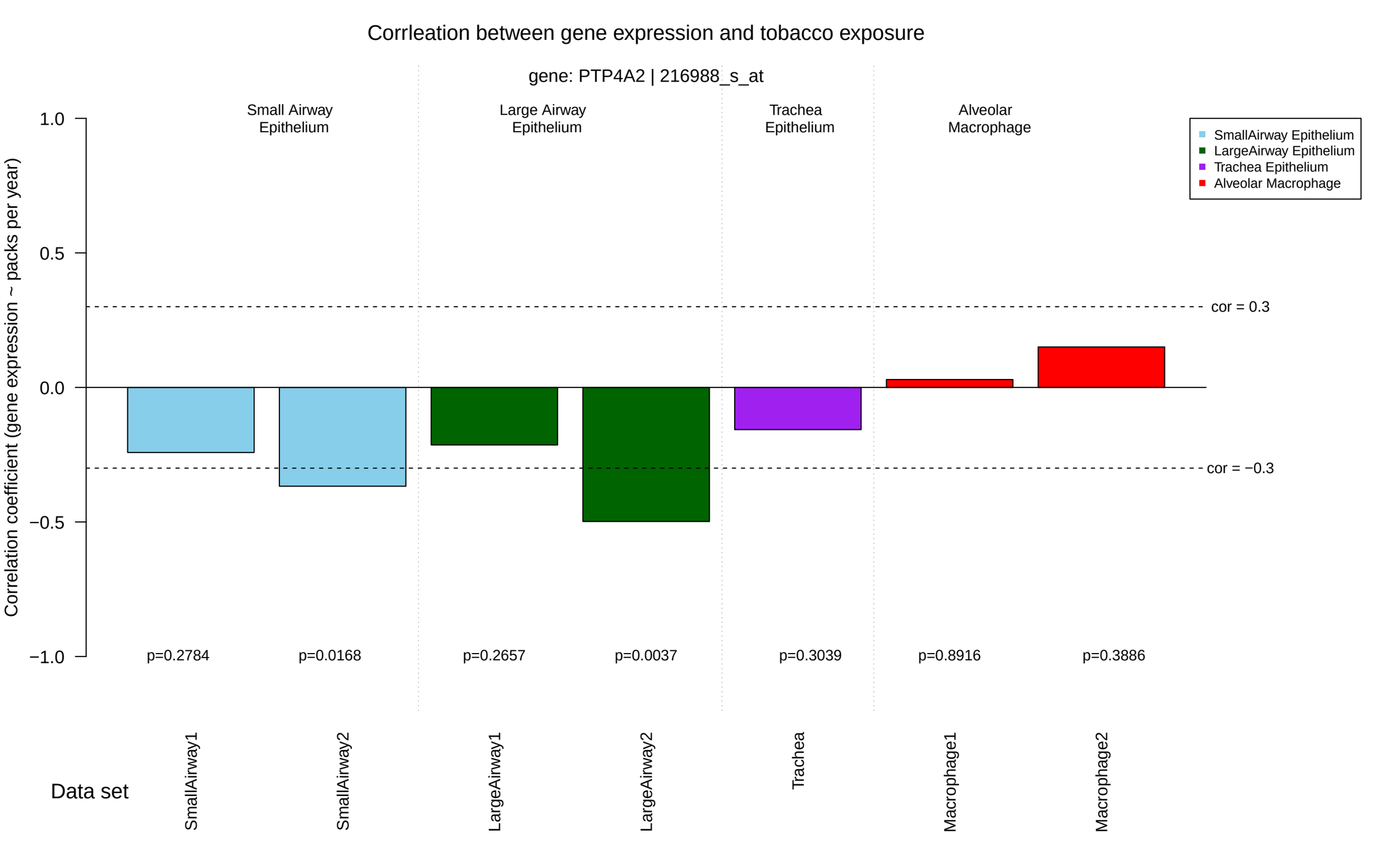

set: LargeAirway1 (p=0.0561)
PTP4A2 | 216988_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1194)
PTP4A2 | 216988_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0662)
PTP4A2 | 216988_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5974)
PTP4A2 | 216988_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2566)
PTP4A2 | 216988_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTP4A2 | 216988_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2784
p=0.0168
p=0.2657
p=0.0037
p=0.3039
p=0.8916
p=0.3886
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2